

Ming Gao

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Employment

Member of Research Staff 2026.07 – now
The Voleon Group, NYC, USA

Education

Ph.D. in Econometrics and Statistics 2021.09 – 2026.06
M.B.A. 2023.09 – 2026.06
Booth School of Business, University of Chicago, Chicago, USA
Advisor: Bryon Aragam

M.Sc. in Statistics 2019.09 – 2021.04
Department of Statistics, University of Chicago, Chicago, USA
Advisor: Jingshu Wang

B.Sc. in Statistics 2015.09 – 2019.06
School of Statistics, Renmin University of China, Beijing, China

Research

11. **Ming Gao**, Wai Ming Tai, and Bryon Aragam. [KL-BSS: Rethinking optimality for neighbourhood selection in structural equation models](#). *Journal of the Royal Statistical Society Series B*, 2026.
10. **Ming Gao**, Yuhao Wang, and Bryon Aragam. [Optimal structure learning and conditional independence testing](#). *International Conference on Machine Learning*, 2026. (Spotlight)
9. **Ming Gao** and Bryon Aragam. [Optimality and computational barriers in variable selection under dependence](#). *Submitted*, 2025.
8. **Ming Gao** and Cong Zhang. [Optimizing return forecasts: A bayesian intermediary asset pricing approach](#). *Submitted*, 2024. (Presented at SFS Cavalcade North America 2025)
7. Jin-Hong Du, Tianyu Chen, **Ming Gao**, and Jingshu Wang. [Joint trajectory inference for single-cell genomics using deep learning with a mixture prior](#). *Proceedings of the National Academy of Sciences*, 2024.
6. Yuhao Wang, **Ming Gao**, Wai Ming Tai, Bryon Aragam, and Arnab Bhattacharyya. [Optimal estimation of gaussian \(poly\)trees](#). *International Conference on Artificial Intelligence and Statistics*, 2024.
5. Yuxuan Guo, **Ming Gao**, and Xiaoling Lu. [Multivariate change point detection for heterogeneous series](#). *Neurocomputing*, 2022.

4. **Ming Gao**, Wai Ming Tai, and Bryon Aragam. [Optimal estimation of gaussian dag models](#). *International Conference on Artificial Intelligence and Statistics*, 2022.
3. **Ming Gao** and Bryon Aragam. [Efficient bayesian network structure learning via local markov boundary search](#). *Advances in Neural Information Processing Systems*, 2021.
2. Goutham Rajendran, Bohdan Kivva, **Ming Gao**, and Bryon Aragam. [Structure learning in polynomial time: Greedy algorithms, bregman information, and exponential families](#). *Advances in Neural Information Processing Systems*, 2021.
1. **Ming Gao**, Yi Ding, and Bryon Aragam. [A polynomial-time algorithm for learning nonparametric causal graphs](#). *Advances in Neural Information Processing Systems*, 2020.

Software

KL-BSS ([Link](#)):

Improving best subset selection (BSS) on neighbourhood selection/variable selection/support recovery in structural equation models and causal discovery.

PC-tree algorithm ([Link](#)):

Optimal estimation of causal poly-tree/forest models via a modified PC algorithm.

NPVAR algorithm ([Link](#)):

Learning nonparametric DAGs in polynomial time with conditional variance assumptions.

TAM algorithm ([Link](#)):

Efficient nonparametric structure learning via Testing and Masking with entropic conditions.

VITAE ([Link](#)):

Single-cell genomics trajectory inference and data integration with uncertainty quantification via variational autoencoders and hierarchical mixture priors.

Teaching Experience

Teaching Assistant

- BUSN 41901: Probability and Statistics (PhD), Autumn, 2023-2024
- BUSN 41916: Bayes, AI and Deep Learning (PhD), Autumn, 2025
- BUSN 41000: Business Statistics (MBA), Winter, 2024-2026

Professional Service

Reviewer

- **Journals:** JASA, JCGS, CSDA, IEEE-TIT, JMLR, TMLR, TNNLS

- **Conferences:** ICML 2023-2026, NeurIPS 2022-2026, AISTATS 2022-2026, ICLR 2025-2026, CLeaR 2025, AAAI 2025, UAI 2025
- **Workshops:** CauScale at AISTATS 2026

Awards

Silver Reviewer	ICML, 2026
Arnold Zellner Doctoral Prize	Chicago Booth, 2024
Stevens Doctoral Program Research Funding	Chicago Booth, 2023 – 2024
Ph.D Program Fellowship	Chicago Booth, 2021 – 2026
Outstanding Thesis Award	RUC, 2019

Internship Experience

Research Intern	2025.06 – 2025.08
The Voleon Group , Berkeley, USA	
Data Analyst Intern	2018.12 – 2019.05
Department of Dynamic Drilling Computing Engineering, Schlumberger , Beijing, China	
Data Analyst Intern	2018.07 – 2018.08
Department of Science and Technology, Industrial Bank of China , Shanghai, China	
Quantitative Research Intern	2018.01 – 2018.03
Lion Fund Management , Beijing, China	